

# **The University of Nottingham Malaysia Campus**

SCHOOL OF COMPUTER SCIENCE

A LEVEL THREE MODULE, SPRING SEMESTER 2023-2024

## **Computer Vision**

Time allowed: TWO HOURS

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*Candidates may complete the front cover of this exam and sign their desk card but must NOT write anything else until the start of the examination period is announced*

**Answer ALL THREE Questions in Section A. Answer TWO out of THREE Questions in Section B.**

*Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.*

*No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.*

**DO NOT turn examination paper over until instructed to do so**

**ADDITIONAL MATERIAL:** Nil.

## SECTION A

Answer ALL THREE Questions in This Section.

**Question 1. The following questions are pertaining to Image Features.**

**[overall 13 marks]**

- a. Briefly describe the main steps to obtain a Histogram of Oriented Gradient (HOG) feature representation for an image patch.

[5 Marks]

- b. Determine the Local Binary Pattern (LBP) string for the central pixel in the image patch in Figure 1.

|     |     |     |
|-----|-----|-----|
| 95  | 100 | 105 |
| 98  | 100 | 105 |
| 103 | 99  | 110 |

Figure 1

[2 Marks]

- c. In Scale Invariant Feature Transform (SIFT), Key Point detection is applied after scale invariant points are obtained. The derivatives of image pixels along the x-axis (horizontal axis) and the y-axis (vertical axis) of a  $3 \times 3$  image patch are shown in Figure 2.

- i. Calculate the Hessian matrix (as explained in the lecture) that is used in Key Point detection. Show your steps but you do not have to provide the final results (e.g., you can show your calculation as  $900 + 600$  instead of 1500)

[4 Marks]

|    |    |   |
|----|----|---|
| 0  | 5  | 1 |
| 2  | 5  | 2 |
| -3 | -2 | 1 |

Pixel derivative along x-axis

|   |    |    |
|---|----|----|
| 3 | 0  | -1 |
| 5 | -1 | 0  |
| 1 | 1  | 5  |

Pixel derivative along y-axis

Figure 2

- [2 Marks]

**[overall 13 marks]**

- [7 Marks]

- $$p^T \mathbf{F} p' = 0 \quad (1)$$

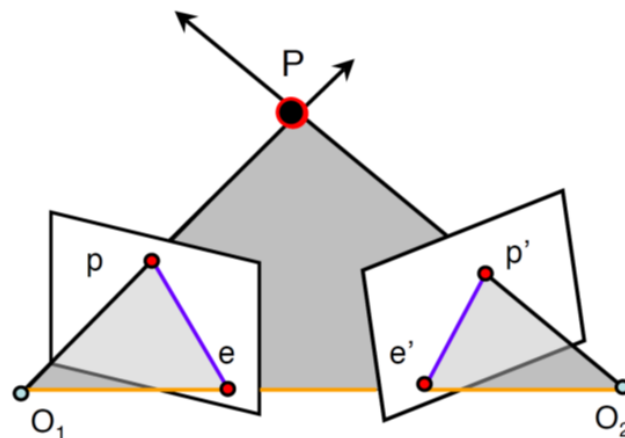


Figure 3

- [2 Marks]

- [4 Marks]

**[overall 14 marks]**

- a. Given a 16-channel input feature map of size  $64 \times 64$ , what is the size of the output feature map if you convolve it with a  $7 \times 7$  filter, without zero padding, and with stride 2, to obtain a 64-channel output feature map? Show your work.

[2 Marks]

- b. Both the operations of 2D convolution and 2D pooling apply filters to a feature map. Explain the differences between a convolution operation and a pooling operation.

[4 Marks]

- c. The model architecture of a Vision Transformer (ViT) for classification which was introduced in the lecture is shown in Figure 4.

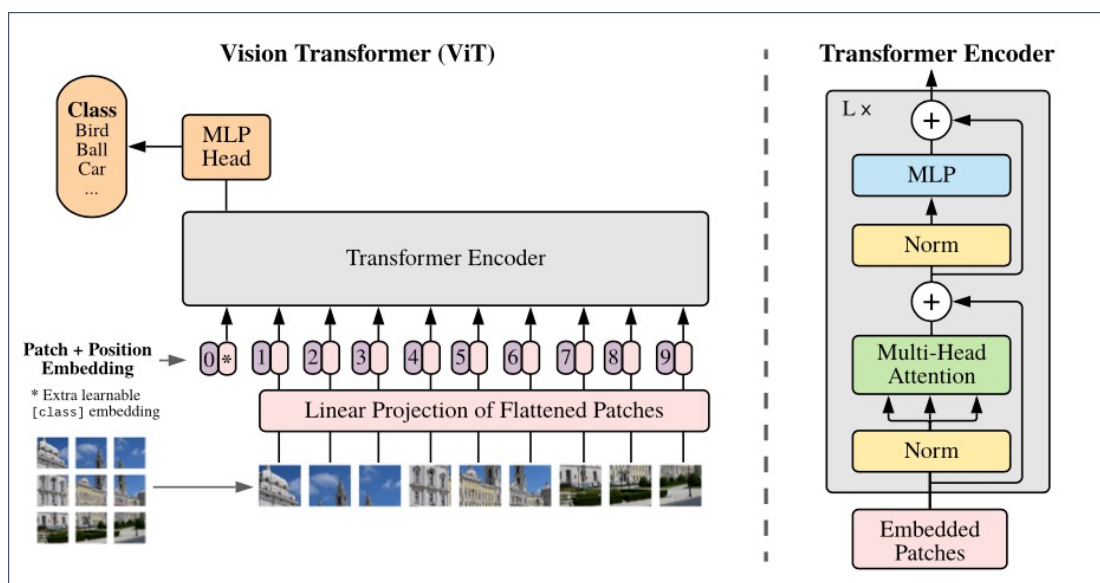


Figure 4

- i. Briefly explain the concept of attention mechanism and what a multi-head attention module does in the transformer encoder block.

[4 Marks]

- ii. Briefly explain what a class token is, its purpose, and how it works for classification task.

[4 Marks]

## SECTION B

Answer TWO Out of THREE Questions in This Section.

**Question 4: The following questions are pertaining to Image Segmentation and Optical Flow.**

**[overall 30 marks]**

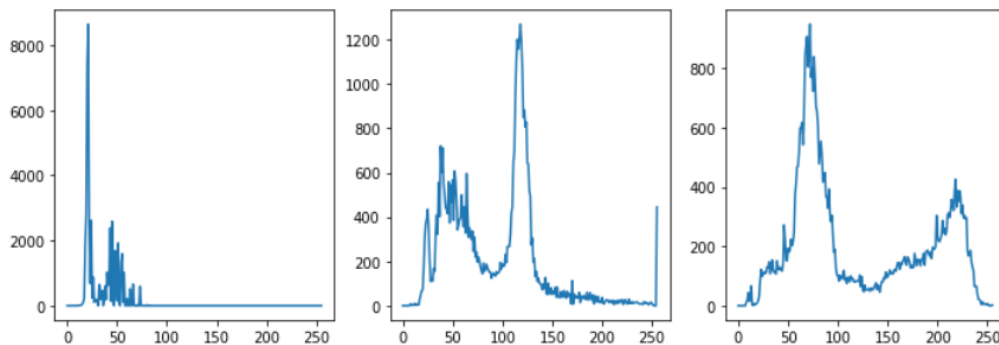
- a. An image in nature is shown in Figure 5. The Hue, Saturation, and Value (HSV) channels of the image are shown in Figure 6a from left to right, and the histograms of the three channels are shown in Figure 6b, respectively. You are required to use a Gaussian Mixture Model (GMM) to segment the image into three regions, i.e., reindeer, grass, and woods, based on the HSV representation of the image. Describe, justify and discuss your design with reference to both histograms and HSV channels.



Figure 5



(a) HSV



(b) Histograms of HSV channels

Figure 6

**[16 Marks]**

- b. Two  $5 \times 5$  image patches at time  $t$  and  $t + 1$  are given in Figure 7(a) and (b), respectively. Calculate the image gradients along  $x, y$ , and the time  $t$  directions for a  $3 \times 3$  window centered at the central pixel of the image patch in (a). Assuming the origin of image coordinate system is at the upper-left corner of the image, use the formula  $I_x = I(x + 1, y) - I(x, y)$  and  $I_y = I(x, y + 1) - I(x, y)$  to calculate the image gradients along  $x, y$  axis, respectively. Based on Lucas-kanade method, explain how these image gradients may be combined to estimate the optical flow vector of the pixel at the centre of the image patch in (a). There is no need to calculate the optical flow vector.

|     |     |     |     |     |
|-----|-----|-----|-----|-----|
| 215 | 214 | 214 | 214 | 214 |
| 215 | 214 | 214 | 214 | 214 |
| 215 | 215 | 215 | 215 | 214 |
| 215 | 215 | 215 | 215 | 213 |
| 215 | 214 | 214 | 214 | 213 |

(a)

|     |     |     |     |     |
|-----|-----|-----|-----|-----|
| 201 | 201 | 200 | 201 | 212 |
| 201 | 201 | 196 | 196 | 207 |
| 201 | 200 | 195 | 195 | 200 |
| 201 | 198 | 192 | 193 | 193 |
| 201 | 196 | 196 | 189 | 191 |

(b)

Figure 7

[14 Marks]

### Question 5: The following questions are pertaining to Stereo Vision

[overall 30 marks]

- a. A pair of rectified stereo images are shown in Figure 8 for the application of a system for inferring package volumetric dimensions. Four pairs of pixels in both left and right images are detected as corresponding pixels and are labelled as "Origin" (red colour), "108 cm" (green colour), "139 cm" (green colour), and "159 cm" (green colour), respectively, which form the three dimensions of the cargo. Explain how the estimation of the three dimensions may be achieved with the stereo images. Include in your answer concepts from the lectures on stereo vision.

[18 Marks]



Figure 8

- b. Figure 9 shows a typical architecture of deep learning based stereo vision model which mimics the traditional stereo vision pipeline. Briefly explain the functionality of each block (A) dense/sparse feature extraction, (B) feature matching, (C) cost volume regularisation, (D) obtain raw disparity map, and (E) post processing and refinement, respectively.

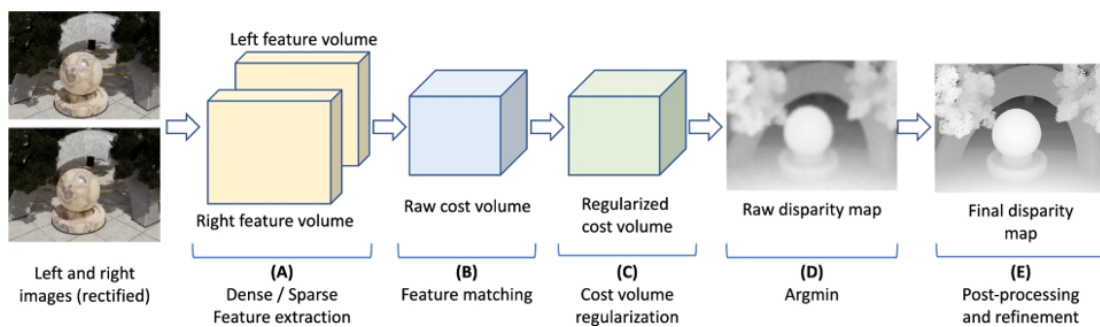


Figure 9

[12 Marks]

**Question 6: The following questions are pertaining to Object Recognition and Deep Learning.**

[overall 30 marks]

- a. The Viola-Jones method relies on efficient integral images to generate a large amount of simple image features. Given the  $3 \times 3$  image patch shown in Figure 10, calculate its integral image, and use this to calculate the simple feature based on subtracting the light-gray region from the dark-gray region. Explain how such pattern can be used to generate multiple simple features in the given image patch.

[10 Marks]

|    |     |     |
|----|-----|-----|
| 90 | 95  | 91  |
| 85 | 120 | 100 |
| 50 | 70  | 60  |

Figure 10

- b. Figure 11 shows a three-dimensional (3D) model reconstruction framework with a Generative Adversarial Network (GAN). Describe the concept of GAN and explain how it is applied in the framework in Figure 11 for the training of and inference with a network that predicts the 3D reconstruction of a 2D input image. Include in your answer concepts from the lectures on GANs.

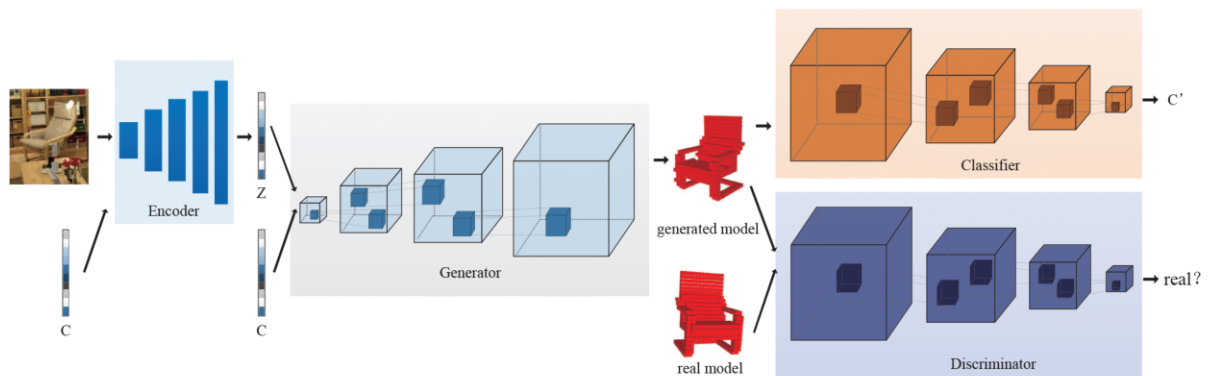


Figure 11

[20 Marks]